

Burakhan Kokcu

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Education

North Carolina State University

Bachelor of Science in Electrical Engineering | GPA 3.1

Bachelor of Science in Computer Engineering | GPA 3.1

Aug. 2022 - Dec 2025
Raleigh, NC

- **Relevant Coursework:** Embedded Systems, Electronics Engineering, Control Systems, Power Control, Complex Digital System Design, Computer Networking, Machine Learning, Logic Design, C++/ C Programming & Software Tools
- **Honors Achievements:** Dean's Honor Roll, Sigma Alpha Pi, Turkish Student Association

Experience

North Carolina State University | Interactive STEM Education Platform

Aug. 2024- April 2025 | Raleigh, NC

- Designed and developed an interactive STEM platform to teach computer logic to 5th–9th grade students, integrating hardware and software concepts.
- Engineered and fabricated custom PCBs with multiple hardware modules, integrating a Raspberry Pi Pico and display for real-time system control and visual feedback.
- Conducted user testing with local educational organizations (e.g., NC Science House) and translated feedback into product improvements and instructional materials, including a structured user guide.
- Authored a structured user guide to improve usability and enhance learning outcomes for target users.

North Carolina State University | Smart Car Developer

Jan 2024 - June 2024 | Raleigh, NC

- Developed a Wi-Fi-enabled remote-controlled vehicle using C, MSP430, and ESP32, enabling real-time command execution and control.
- Designed modular hardware architecture (motor, sensor, and display subsystems) and integrated IoT communication for improved responsiveness and flexibility.
- Applied finite state machine (FSM) design patterns to manage system behaviors such as navigation, communication, and user interaction. Assembled and soldered multi-board hardware system, ensuring reliable electrical connections and system stability.

Academic/ Personal Projects

Multi-Agent AI Content Analysis Platform | Python, streamlit

Jan 2025 - April 2025

- Developed a web-based platform leveraging multi-agent AI systems to perform automated content analysis and annotation.
- Designed system architecture to support scalable, automated qualitative analysis with minimal human intervention.

Portfolio Website | React, JavaScript, HTML

Mar 2026 - May 2026

- Deployed a robust personal portfolio website utilizing a modern tech stack JavaScript, HTML, Tailwind, React.js.
- Implemented responsive design for mobile access, significantly boosting user visits.
- Demonstrated strong initiative and technical proficiency in front-end development and deployment practices.

Engineering Strengths

- Hands-on prototyping and hardware development
- Strong debugging and system integration skills
- Comfortable working in lab and workshop environments
- Ability to translate concepts into working systems

Technical Skills

Developer Tools & Technologies:: Fusion 360, AutoCAD, Wireshark, AWR, PSpice, PCB Design Tools

Hardware & Engineering: Circuit Design, PCB Design, Embedded Systems

Languages: Python, C, C++, Java, MATLAB, HTML, VHDL, Verilog

Design & Modeling: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems.